Enhancing Catalyst Synergy by Integration with Perovskite-Based Support Bifunctional Catalysts

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Introduction

Designing and controlling synergistic effects between two chemocatalytically active sites is a current challenge. By combining catalysts, distinct consecutive reaction pathway and selectivity in coupled catalytic reactions could be achieved[1]. We focus on integrating the dehydrogenation of propane (DHP) and the reversible water gas shift reaction (rWGS). By combining these reactions with a hydrogen storage support, the large energy required for releasing hydrogen to the atmosphere can be offset by the energy used for adsorbing hydrogen as shown in Figure 1.

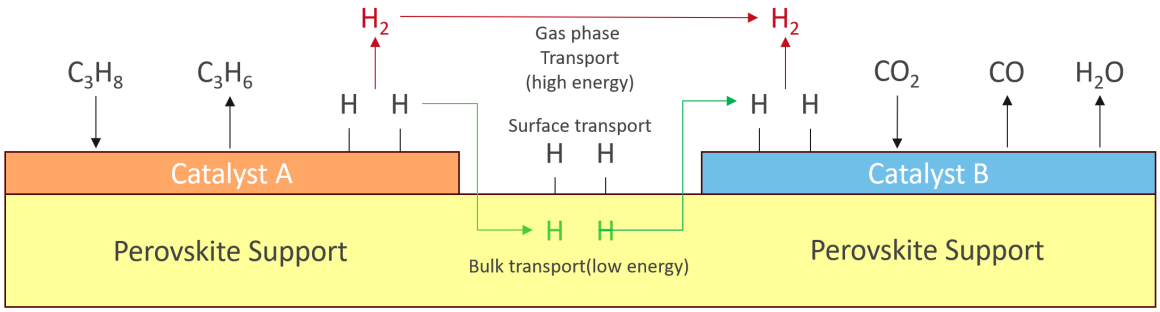

Figure 1 Two possible routes for the reactant transport between two supported active catalysts: hydrogen produced on catalyst A can be transported through gas phase with high energy or through support surface and bulk with low energy.

The catalyst support plays a crucial role in modulating the geometric and electronic structure of active metal phases for optimizing the catalytic activity, selectivity and stability. $BaZrO_3$ (BZO) are of particular interest due to its high proton conductivity[2], low thermal conductivity, and chemical inertness. Therefore, I propose to enhance the synergy between consecutive reactions by designing DHP and rWGS catalysts, BZO as support material.

Materials and Methods

We examined the hydrogen binding energy in the perovskite support and on its (001) surface using Density functional theory (DFT) for a variety of dopants. The DHP catalyst was prepared, analyzed and applied. In the cooperation research with Bascat, (M)-γ-Al_2O_3 (M=Ni, Mg, Cu, Zn) prepared by sol-gel process was tested as

the DHP catalyst. The BZO support was prepared by self-combustion. We prepared and examined the composite Al_2O_3/BZO material by a modified sol-gel route.

Results and Discussion

According to DFT calculation, Yttrium and Indium doped BZO shows a promising hydrogen binding energy which favors the reversible adsorption and desorption process. For the surface, different yttrium doping sites were compared according to Bader Charge Analysis. With above results, we further analyzed the segregation energy and built up the energy profile for hydrogen migration process.

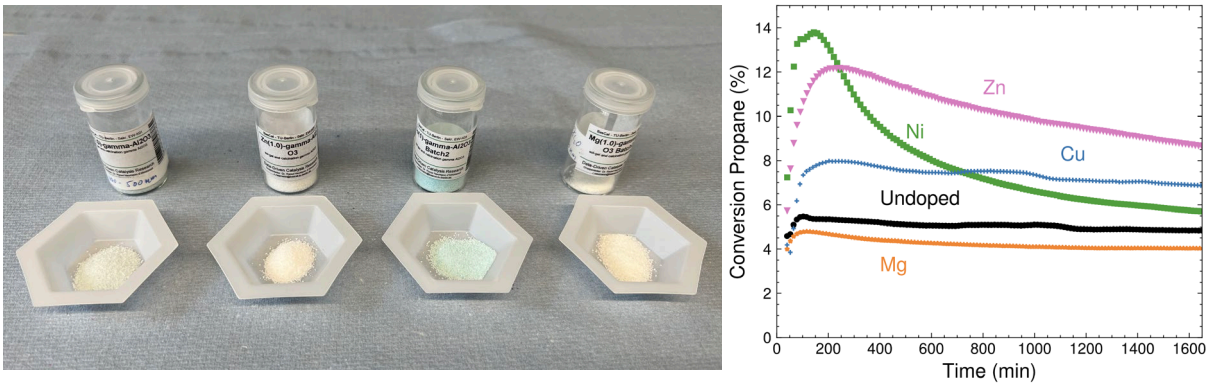

Figure 2 Left: γ-Al_2O_3 doped with different metals: Ni, Zn, Cu, Mg (left to right). Right: Propane conversion of different metal doped γ-Al_2O_3.

For the DHP catalyst, we examined the influence of doping with transition metals on the catalytic activity, stability and selectivity. The calcination temperature was optimized to 800°C and the doping concentration was optimized to 1:155 (M:Al atom ratio). Zn doped Al_2O_3 shows higher conversion and less coking as shown in Figure 2.

Significance

We explored and explained the connection between the DFT simulation and the experimental results. Furthermore, to control the support perovskite structure by varying the experimental conditions to closely reflect the DFT-optimized structures. We propose that future work will focus on the rWGS catalyst and the integration method for these two catalysts and the support perovskite. Besides, application of the integrated catalyst in electrochemical reactions will be conducted.

References

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